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Treatment options of nitrogen heterocyclic compounds in industrial wastewater: From fundamental technologies to energy valorization applications and future process design strategies

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ABSTRACT

Nitrogen heterocyclic compounds (NHCs) widely exist in industrial wastewater and presented significant environmental and health risks due to their toxicity and persistence. This review addressed the challenges in treating NHCs in industrial wastewater, focusing on developing sustainable and efficient treatment processes. While various technologies, including adsorption, advanced oxidation/reduction processes (AOPs/ARPs), and microbial treatments, have been studied at the experimental stage of treating synthetic wastewater, scale-up for industrial applications is imperative. After analyzing the characteristics of NHCs and evaluating different treatment methods with the aid of efficiency and cost-benefit analysis, efficient detoxification while maximizing energy recovery constitutes a critical requirement in treating NHC-containing wastewater. Hence, we proposed a comprehensive strategy combining hydrolysis-acidification pretreatment enhanced by electro-assisted microaeration with methanogenic anaerobic digestion as core treatment units. The process design for NHC-containing wastewater treatment should consider the dynamic balance between removal efficiency, energy consumption, and ammonia recovery, incorporating environmental and economic impacts through life cycle assessment and technical-economic analysis. The potential of machine learning in optimizing operational parameters, predicting effluent quality, and supporting process design decisions is promising. To develop interpretable and practical solutions, the integration of data-driven approaches with mechanistic understanding and prior knowledge is indispensable. This review provided novel insights into sustainable NHC treatment strategies in the context of energy valorization and artificial intelligence advancement, offering guidance for future research and industrial applications.

1. Introduction

Nitrogen heterocyclic compounds (NHCs) are a class of organic compounds characterized by a ring structure containing one or more nitrogen atoms in addition to carbon atoms. The simplest structure is three-membered ring (e.g., aziridine), while the most stable and prevalent forms are five-membered ring (e.g., pyrrole), six-membered ring (e.g., pyridine) and fused-ring (e.g., quinoline and indole). NHCs widely exist as raw materials, intermediates, and products in industries such as pharmaceuticals, food processing, refining, and chemical engineering, and are present in the wastewater generated by these sectors (Ciarlini et al., 2024; Gong et al., 2016; Hu et al., 2019; Jia et al., 2024; Shi et al., 2019d; Yang et al., 2024b). Most NHCs pose toxicity and carcinogenic risks, causing significant harm to the environment and human health (Shi et al., 2019a). Therefore, removing NHCs from wastewater to mitigate their associated risks is crucial.

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Various fundamental technologies have been employed to remove NHCs from (synthetic) industrial wastewater, including adsorption, advanced oxidation/reduction processes (AOPs/ARPs), and microbial treatment. Adsorbents typically feature complex pore structures and large specific surface areas, such as carbon-based materials (Cui et al., 2020), metal-organic frameworks (MOFs) (Shin et al., 2019), and synthetic resins (Sun et al., 2022). While many of these adsorbents are readily available and cost-efficient, adsorption merely transfers NHCs from the aqueous phase to the surface or pores of the adsorbent without achieving degradation or mineralization. Additionally, practical challenges such as adsorption saturation and difficulty in recovery limit the expansibility of this method. AOPs and ARPs generate highly reactive species, such as hydroxyl radicals (•OH), superoxide ($\bullet O_2^-$) and H• radicals, which rapidly degrade and mineralize NHCs. These processes can be categorized based on the radical generation method, including Fenton (Ben Hammouda et al., 2016), electrochemical (Moreira et al., 2017), ozone-catalytic (Liu et al., 2022a), photocatalytic (Gupta et al., 2020), etc. However, AOPs and ARPs are often associated with high costs and energy consumption, limiting their promotion and application in wastewater treatment.

Microbial treatments are more cost-effective and widely applied compared to physicochemical methods. For instance, in aerobic conditions, monooxygenation and ammonification reactions convert NHCs into hydroxylated intermediates and NH_4^+, which can be further oxidized to inorganic carbon and nitrogen (Xu et al., 2017); in anaerobic conditions, NHCs can be converted into biogas, enabling energy recovery. However, microbial treatments have some drawbacks, including high aeration energy consumption in aerobic systems and methanogenesis inhibition in anaerobic systems. As a result, many researchers focus on the development of bioaugmentation processes. For example, the use of granular sludge in aerobic systems (Tomar et al., 2022) or the construction of anoxic-aerobic reactors (Hou et al., 2018; Shen et al., 2015), which utilize NO_3^- produced in the outer layer/aerobic zone as an electron acceptor to drive the degradation of NHCs in the inner layer/-anoxic zone. Strategies such as adding co-digestion substances (Shi et al., 2020a), introducing carbon-based/iron-based conductive materials (Li et al., 2019; Yang et al., 2024b), or applying external voltage (Gao et al., 2020) have been employed to enhance anaerobic bioconversion, mineralization, and methane production. The addition of specific functional bacteria or enzymes can also enhance the biodegradation of NHCs (Jia et al., 2024; Niu et al., 2023).

Current research on adsorption, AOPs, and microbial treatment of NHCs primarily remains at the experimental stage, lacking efficient processes suitable for industrial-scale implementation. Moreover, existing process designs have not systematically considered the integration of removal efficiency, energy and ammonia recovery potential, and environmental and economic impacts. This review provided a detailed overview of the characteristics of NHCs and current progress in various physicochemical and microbial treatment options, as well as strategies for their enhancement. NHC removal performances of different treatment methods were discussed with the aid of efficiency and cost-benefit analysis. Additionally, this review attempted to propose comprehensive strategies for optimizing core treatment units capable of efficient detoxification and energy valorization, and designing adaptable and scalable treatment processes. Future perspectives and research gaps in energy-resource recovery from industrial wastewater and systematical process design were also outlined. This review aims to inspire pioneering research on NHCs degradation and conversion in the context of sustainability and artificial intelligence advancement.

2. Sources and toxicity of NHCs

2.1. Natural sources of NHCs

NHCs are abundant in nature as structural components of diverse biological compounds (Obaid et al., 2022). In nucleic acids, they form pyrimidine and purine base pairs of RNA and DNA. Essential vitamins containing NHC structures include Vitamin B1 (pyrimidine and thiazole rings), B2 (pteridine ring), B3 (pyridine ring), B6 (pyridine ring), B12 (reduced pyrrole rings), and folic acid (pteridine ring). Three amino acids - proline (pyrrole ring), tryptophan (indole ring), and histidine (imidazole ring) - also incorporate NHC structures (Joule, 2016). Additionally, NHCs are integral to electron carriers ($NAD(P)^+$/NAD(P)H), flavin nucleotides (FAD/FADH), and energy storage molecules (ATP and GTP).

Pyridine derivatives are naturally present in various plants and foods, including henbane, tung oil, and coffee. Indole derivatives, comprising over 3000 natural compounds distributed across diverse flora (especially in cruciferous vegetables), exist extensively in proteins, amino acids, and biologically active alkaloids, with more than 40 compounds serving therapeutic roles in diabetes, cancer, and asthma treatment (Kim et al., 2022; Zhang et al., 2024). Quinoline derivatives predominantly exist in nature as quinoline alkaloids, which exhibit broad distribution across plant species. For instance, quinine, camptothecin, and dictamnine could be isolated from *Cinchona* plant, *Camptotheca acuminata*, and *Dictamnus albus*, respectively. Quinoline alkaloids demonstrate significant pharmaceutical potential through their antimicrobial, anticancer, antimalarial, and anti-tuberculosis properties, making them valuable in drug development (Afzal et al., 2015; Hu et al., 2017; Lauria et al., 2021).

2.2. Anthropogenic sources of NHCs

Industrial sources of NHCs primarily stem from energy, pharmaceutical, chemical, and food sectors. Wastewater treatment facilities primarily manage NHCs generated from these human production activities (Table S1).

In the coal processing industry, three main processes - carbonization, gasification, and liquefaction - generate NHC-containing waste streams (Shi et al., 2019c). Coal gasification wastewater presents a particularly severe environmental challenge, with NHCs constituting 60 % of its toxic, refractory organic compounds, predominantly comprising pyridine, indole, quinoline, and their derivatives (Shi et al., 2018). Similarly, shale oil production generates wastewater characterized by high chemical oxygen demand (COD), elevated kjeldahl nitrogen levels, hydrocarbons, and various heterocyclic compounds including pyrrole, pyridine and its methylated derivatives (2,6-dimethyl-, 2,4,6-trimethyl-, 2-dimethylamino-), quinoline, and isoquinoline (Hawthorne and Sievers, 1984).

Pharmaceutical manufacturing processes generate high-concentration NHC wastewater during the production of various therapeutic compounds. Notable examples include nicotinamide (vitamin B complex), isoniazid (anti-tuberculosis), diethylnicotinamide (central stimulant), tropisetron (antiemetic), cetylpyridinium chloride (antiseptic), and esomeprazole (gastroesophageal reflux treatment) (Padoley et al., 2008). For instance, nicotinamide production wastewater contains approximately 5000 mg/L nicotinamide, 3000 mg/L nicotinic acid, and 1000 mg/L 3-cyanopyridine.

The food industry generates NHCs through Maillard reactions (condensation between amino moieties and carbonyl groups of reducing sugars), which are widely employed in baking, meat and coffee processing, flavoring production, tobacco manufacturing, and beverage brewing to enhance product characteristics (Lund and Ray, 2017).

2.3. Toxicity of NHCs

NHCs are extensively utilized in energy, pharmaceutical, chemical, and dye industries. Their discharge poses health risks due to their carcinogenic, teratogenic, and mutagenic properties (Shi et al., 2019a). Notable examples include quinoline, a highly mobile and refractory pollutant (LD50: 460 mg/kg) (Liu et al., 2016), which causes mucosal damage (eyes, nose, and throat) through vapor exposure; pyridine,

classified as a Group 2B carcinogen by the World Health Organization, which induces mucosal and visceral organ damage upon prolonged exposure and exhibited strong irritating odor (Zhao et al., 2023); and indole, which demonstrates both acute and chronic toxicity (He et al., 2022), potentially leading to cardiovascular dysfunction, arterial hypertension, and coronary heart disease. Moreover, the resulting heterocyclic amines through Maillard reactions, classified as NHCs, often fall under category 2A or 2B carcinogens (Bellamri et al., 2023; Olalekan Adeyeye and Ashaolu, 2021).

NHCs discharged into water bodies present heightened environmental risks compared to their homocyclic analogues due to their superior solubility, more bioavailable and tendency for environmental persistence (Shi et al., 2024; Bartoš et al., 2006). Their ecotoxicological impact is particularly severe on aquatic and soil organisms. Specifically, acridine and carbazole exhibit acute toxicity to zebrafish embryos with LC50 values of 0.7 and 1.1 mg/L, respectively, causing severe developmental impairments (Peddinghaus et al., 2012), while 7H-benzo[c] carbazole demonstrates high toxicity to freshwater algae *Raphidocelis subcapitata* with an EC50 of 0.25 mg/L (Çelik et al., 2023). Pyridine disrupts *Eisenia fetida* homeostasis by inducing excessive reactive oxygen species production and compromising coelomocyte membrane integrity, leading to cascading detrimental effects (He et al., 2023).

As for their genotoxicity, the planar fused-ring structures of NHCs enable their intercalation between DNA bases. During biotransformation, highly electrophilic intermediates, particularly diol epoxides, form reversible but strong bonds with nucleophilic groups on nucleic acids and proteins (e.g., carboxyl, amino, and sulfhydryl groups). The subsequent formation of covalent bonds between carbonium ion intermediates and DNA helices generates DNA adducts, potentially disrupting DNA configuration and initiating mutagenesis and carcinogenesis (Bartoš et al., 2006). Bartoš et al. (2006) conducted SOS Chromotest (prokaryotic genotoxicity) and green fluorescent protein test (eukaryotic genotoxicity) on 20 NHCs (mainly polyaromatic NHCs), and found that NHCs have higher genotoxic potentials than their homocyclic analogues. Gabelova (2020) reviewed that 7H-Dibenzo[c,g]carbazole (a common polyaromatic NHC) induces DNA adducts in fish and human fibroblasts, while also causing mutations in diploid human fibroblasts and micronuclei formation in human blood lymphocytes.

3. Adsorption removal of NHCs

The adsorbents for NHCs primarily consisted of three types of adsorptive materials: carbon-based adsorbents, metal-organic frameworks (MOFs), and synthetic resins. They removed NHCs through π-complexation, van der Waals forces, acid-base interactions, etc. (Fig. 1). Table S2 summarized the operational conditions and adsorption performances of these materials for NHCs.

3.1. Carbon-based adsorbents

Activated carbon and carbon nanospheres exhibited strong adsorption capacities for NHCs, as reported by Liao et al. (2013), Liao et al. (2012) and Rameshraja et al. (2012), because of their well-developed porous structure, large specific surface area, stable physicochemical properties, and amenability to chemical modification. This adsorption process was endothermic ($\Delta H>0$), entropy-increasing ($\Delta S>0$), and spontaneous ($\Delta G<0$). The predominant mechanism for NHCs chemical adsorption on activated carbon (including granular activated carbon (GAC) and biochar) was π-π conjugation interaction. The adsorption rate in the early stage (e.g., within 1 h) was determined by surface diffusion, followed by pore diffusion as the rate-limiting step. Both GAC and biochar showed excellent adsorption performance: GAC achieved an adsorption capacity of 160–175 mg/g for quinoline (at 15–45 °C) (Rameshraja et al., 2012), while biochar (bamboo charcoal)

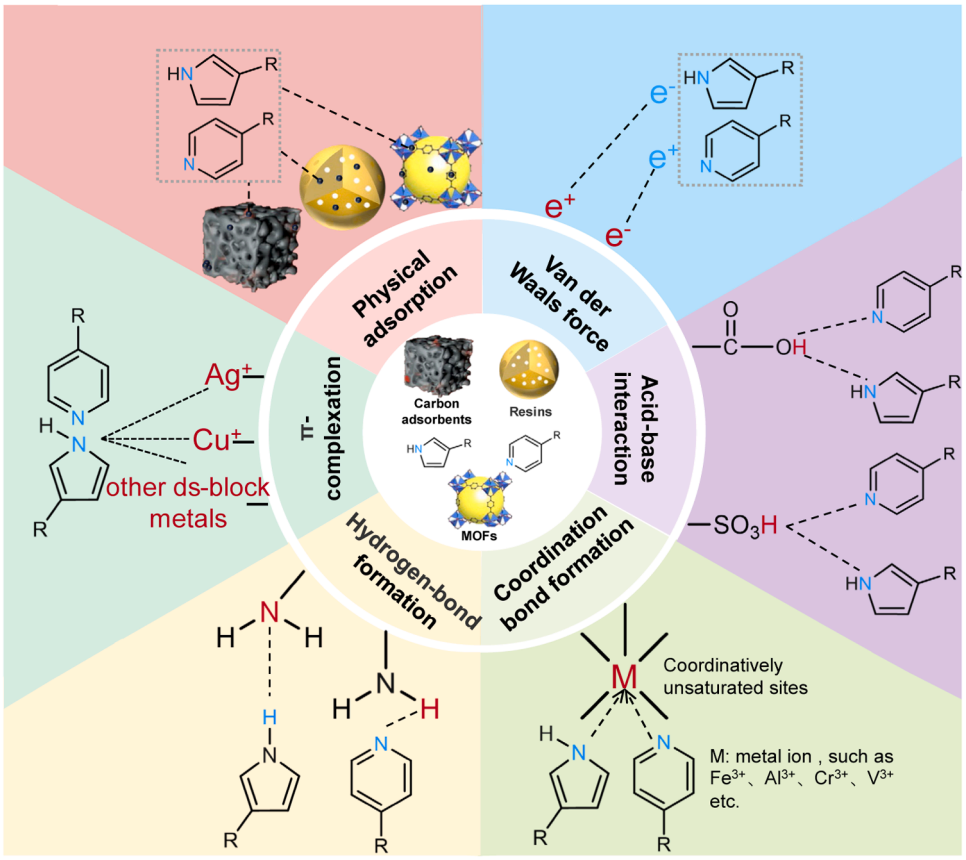


Fig. 1. Mechanisms of NHCs removal by carbon-based adsorbents, metal-organic frameworks, and hypercrosslinked resins.

demonstrated good affinity for pyridine, indole, and quinoline (Liao et al., 2013). As for carbon nanospheres, they could be doped with Fe_3O_4 magnetic nanoparticles, enabling easy recovery through external magnetic fields while maintaining their excellent adsorption properties (Cao et al., 2016; Wang et al., 2016b).

Surface modification of carbon-based adsorbents could be achieved through the interaction between unsaturated bonds and heteroatoms (O, N, S), resulting in various functional groups that determine their adsorption properties. The selection of appropriate modification methods was crucial for enhancing their performance. Alkali modification was an effective and practical method to enhance carbon-based adsorbents by simultaneously improving their physical and chemical properties. This treatment increased pore volume and specific surface area while introducing acidic oxygen-containing functional groups, thereby strengthening both porous adsorption and acid-base interactions with Lewis basic NHCs such as pyridine, indole, and quinoline. For instance, NaOH-modified activated carbon (bamboo charcoal) showed increased NHCs adsorption capacity (Liao et al., 2012); KOH modification could enhance the quinoline adsorption performance of magnetic carbon nanospheres (Cui et al., 2020). In contrast, acid modification by H_2SO_4 or HNO_3 only increased the number of acidic functional groups on the surface, but reduced the pore volume and specific surface area (de Oliveira Ferreira et al., 2019), resulting in less effective adsorption compared to alkali modification (Liao et al., 2012). Microwave radiation served as an efficient reagent-free modification method for carbon-based adsorbents, improving their performance through dual mechanisms: the expansion of pore size and specific surface area for improved porous adsorption, and the enhancement of hydrophobicity for stronger π-π conjugation interactions with NHCs (Liao et al., 2013; Liao et al., 2012). The incorporation of transition metal ions (Cu^+, Ag^+, Pd^{2+}) into carbon-based adsorbents enhanced NHCs removal through π-complexation interactions, with benefits that outweighed the slight reduction in specific surface area and pore volume. For instance, CuCl@activated carbon, synthesized through Na_2CO_3 reduction of $CuCl_2$@activated carbon at room temperature, showed doubled adsorption capacity for quinoline-indole mixtures versus unmodified activated carbon (Ahmed and Jhung, 2015).

Nitrogen-doped porous carbons (NDPCs) featured a large specific surface area, well-developed pore structures (micropores, mesopores, and macropores), abundant functional groups, and nitrogen dopants (pyrrolic N, pyridinic N, and quaternary N). The presence of mesopores and macropores allowed NHC molecules to rapidly transfer from the external environment into the porous carbon material, where they were fixed through π-π conjugation interactions with the inner walls of the micropores. Surface modification through acid-base treatment generated diverse functional groups that strengthened acid-base interactions between NDPCs and NHCs. Additionally, nitrogen doping, especially pyrrolic N sites, enhanced adsorption heat, providing stronger binding sites for NHC molecules. For instance, Xue et al. (2019) prepared a kind of NDPC by carbonizing a precursor, which was synthesized by reacting *2,4,6-tris*(chloromethyl)mesitylene with ethylenediamine, under 700 °C. This NDPC-700 proved an indole adsorption capacity as high as 861 mg/g.

3.2. Metal-organic frameworks (MOFs)

The main mechanisms of NHCs adsorption by MOFs included acid-base interaction, π-complexation, Van der Waals forces, and hydrogen bonding interactions (Ahmed and Jhung, 2016a). Most MOFs functioned as Lewis acids through their coordinatively unsaturated sites (CUSs), which interacted with NHCs acting as Lewis bases (Maes et al., 2011). When MOFs contained d-block metals or their cations, they adsorbed NHCs via π-complexation, with NHCs adopting a planar orientation that minimized steric hindrance and improved adsorption efficiency (Ahmed and Jhung, 2016a). Van der Waals interaction, characterized by low adsorption heat, was a common mechanism for NHC removal by MOFs at ambient temperature (Ahmed et al., 2013). Hydrogen bonding interaction contributed significantly to NHC adsorption on modified MOFs. The presence of both H-acceptors and H-donors on modified MOF surfaces ensured hydrogen bonding with NHCs, regardless of the NHC's hydrogen bonding role. This mechanism was exemplified in MIL-125-VFG, which could form hydrogen bonds with both H-donor indole and H-acceptor quinoline (Ahmed and Jhung, 2017; Shin et al., 2019).

Basic MOFs (UiO-66, MIL-101, MIL-125, and MAF-6) demonstrated restricted NHC adsorption capacities (indole: 127–410 mg/g; quinoline: 103–422 mg/g (Ahmed et al., 2017; Khan et al., 2020; Sarker et al., 2018; Shin et al., 2019)), motivating continued research for performance enhancement. Three typical examples of MOFs modification were: introducing functional groups (-NH-, -C(O)-, and -C(O)-OH) into MOFs, grafting adenine onto the CUSs of MOFs, and combing graphene oxide with MOFs. They could increase the maximum adsorption capacity (20–25 °C) for indole from less than 410 mg/g to 637, 532, and 542–593 mg/g, respectively, while for quinoline from less than 422 mg/g to 546, 511, and 484–498 mg/g, respectively (Ahmed and Jhung, 2016b; Sarker et al., 2018; Shin et al., 2019).

3.3. Hypercrosslinked resins

Hypercrosslinked resins also possessed large specific surface areas, rich micro and mesoporous structures, and served as efficient adsorbents for NHCs adsorption. Common hypercrosslinked resins were synthesized through a post-crosslinking reaction between chloromethylated polystyrene and divinylbenzene (Wang et al., 2010). The main mechanisms of NHCs adsorption by hypercrosslinked resins included π-π conjugation, hydrophobic interactions, and electrostatic attraction (Sun et al., 2022; Zhu et al., 2016). Similar to activated carbon, this adsorption process was endothermic ($\Delta H>0$), entropy-increasing ($\Delta S>0$), and spontaneous ($\Delta G<0$). Nonpolar resins were suitable for adsorbing hydrophobic NHCs (such as indole) from polar solutions through van der Waals forces, while polar resins, typically containing strong polar groups such as sulfonic acid groups, mainly adsorbed NHCs through hydrogen bonding, acid-base interactions, and electrostatic interactions. Sulfonation of hypercrosslinked resins produces divergent effects on NHC adsorption, depending on the NHC structure. For monocyclic NHCs (pyridine, imidazole, pyrazine), increased sulfonation enhances adsorption through electrostatic attraction, doubling the capacity for pyridine compared to unsulfonated resin. However, for polycyclic NHCs (indole, quinoline, benzimidazole), which rely more on hydrophobic interaction and π-π conjugation, sulfonation reduces adsorption efficiency, dropping quinoline capacity to 80 % of the unsulfonated resin's performance (Sun et al., 2022).

3.4. Comparison of operating conditions and adsorption efficiencies

Despite being endothermic, NHC adsorption processes operated effectively at room temperature (25–30 °C), eliminating the need for additional heating energy. While these adsorbent materials functioned across various pH conditions, optimal adsorption occurred near neutral pH (7 ± 2), as highly acidic conditions created positive surface charges that repelled cationic NHCs (weakly alkaline NHCs) (Cui et al., 2020). The reported adsorbate concentrations (Table S2) vary widely, reaching up to 2000, 10,000 and 1000 mg/L for indole, quinoline and pyridine, respectively. Among adsorbents, modified MOFs exhibited superior NHCs uptake compared to carbon-based and resin materials; however, their enhanced adsorption capacity required complex preparation and modification processes. A critical limitation of MOFs is their water stability. Although ethanol-based regeneration can be performed 4–5 times with minimal capacity loss (Ahmed et al., 2017; Khan et al., 2020; Sarker et al., 2018; Shin et al., 2019), even the most thermodynamically stable MOFs (e.g., MIL-101) maintain stability for only weeks in pure water and days in harsher conditions (Burtch et al., 2014), with further

deterioration in complex industrial wastewater matrices. This instability stems from water molecules attacking the coordination bonds within MOF frameworks, leading to structural collapse, necessitating hydrophobic modifications and steric hindrances reinforcement (Ahmed and Jhung, 2017; Yuan et al., 2016). In conclusion, while adsorption offers advantages such as simple application (direct dosing without energy supply or reactor requirements) and broad treatment concentration ranges, concerns remain regarding cost-efficiency, regeneration capacity, and long-term stability.

4. Chemical degradation of NHCs

4.1. Advanced oxidation processes (AOPs)

AOPs generated powerful reactive species through various activation methods including high temperature/pressure, electrocatalysis, photonic irradiation, and catalysts. These processes primarily produced •OH, alongside other reactive species such as $•O^{2-}$, hydroperoxyl ($•HO_2$), and sulfate ($SO_4^{-}•$) radicals, as well as non-radical oxidants such as singlet oxygen ($^1O_2$), H_2O_2, surface activated oxides/ radicals, electron transfer mechanism and high-valent metals. The ability of these reactive species to achieve complete decomposition or mineralization of most NHCs made AOPs a promising treatment technology.

4.1.1. Degradation pathways of NHCs in AOPs

Fig. 2 illustrated the AOP degradation pathways of typical NHCs (pyridine, indole, quinoline, and etc.) (Gong et al., 2016; Liu et al., 2023; Lou et al., 2021; Singh and Lo, 2017; Zhu et al., 2017b). The process initiated with hydroxylation, followed by concurrent carboxylation/amination and heterocyclic ring-opening reactions. These transformations generated intermediates that further broke down into smaller organic molecules, ultimately mineralizing to CO_2, H_2O, and NO_3^-.

4.1.2. Degradation mechanisms of NHCs in AOPs

The AOPs for NHC degradation primarily consisted of Fenton/ Fenton-like, photocatalysis, electrochemical oxidation, catalytic ozone oxidation, etc. (Fig. 3). Table S3 summarized the operational conditions and NHC degradation performances of these AOPs.

(1) Fenton/Fenton-like

The conventional Fenton process operated under acidic conditions, where Fe^{2+}-catalyzed decomposition of H_2O_2 generated •OH with high oxidation potential. This initial radical triggered cascading reactions that produced additional reactive oxygen species (ROS), facilitating the oxidative degradation of NHCs (Fig. 3a). The underlying mechanisms were as follows (Eqs. (1)–(3)) (Liu and Wang, 2023; Padoley et al., 2011):

$$H_2O_2 + Fe^{2+} \rightarrow •OH + OH^- + Fe^{3+},\ k = 40–80\ M^{-1}S^{-1} \quad (1)$$

$$H_2O_2 + Fe^{3+} \rightarrow H^+ + Fe^{2+} + HO_2•,\ k = 0.001–0.01\ M^{-1}S^{-1} \quad (2)$$

$$•OH + HO_2• + NHCs \rightarrow CO_2 + H_2O + NO_3^- \quad (3)$$

Heterogeneous catalysis has evolved to address the iron sludge limitations of traditional free Fe^{2+} systems. Iron alginate beads (Fe-ABS) effectively removed indole (100 % removal of 20mg/L and 72 % TOC

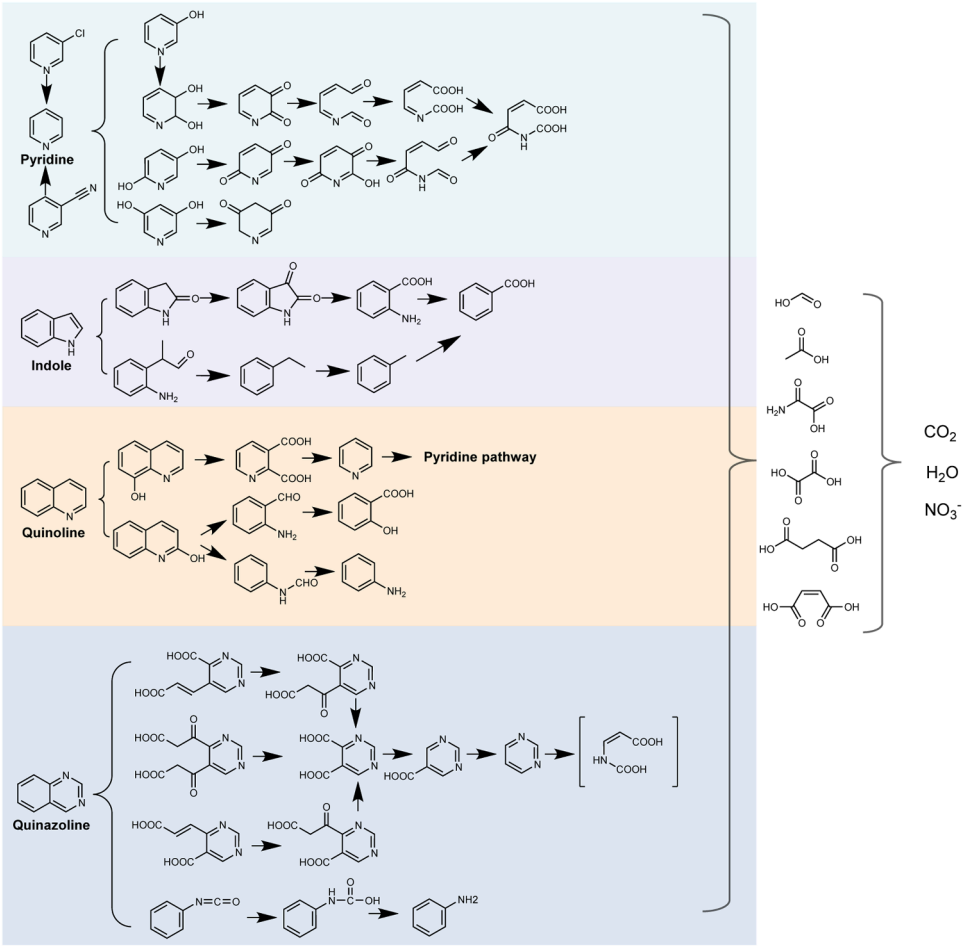


Fig. 2. The degradation pathway of typical NHCs oxidized by •OH, $•O_2^-$, and $•HO_2$ during advanced oxidation processes.

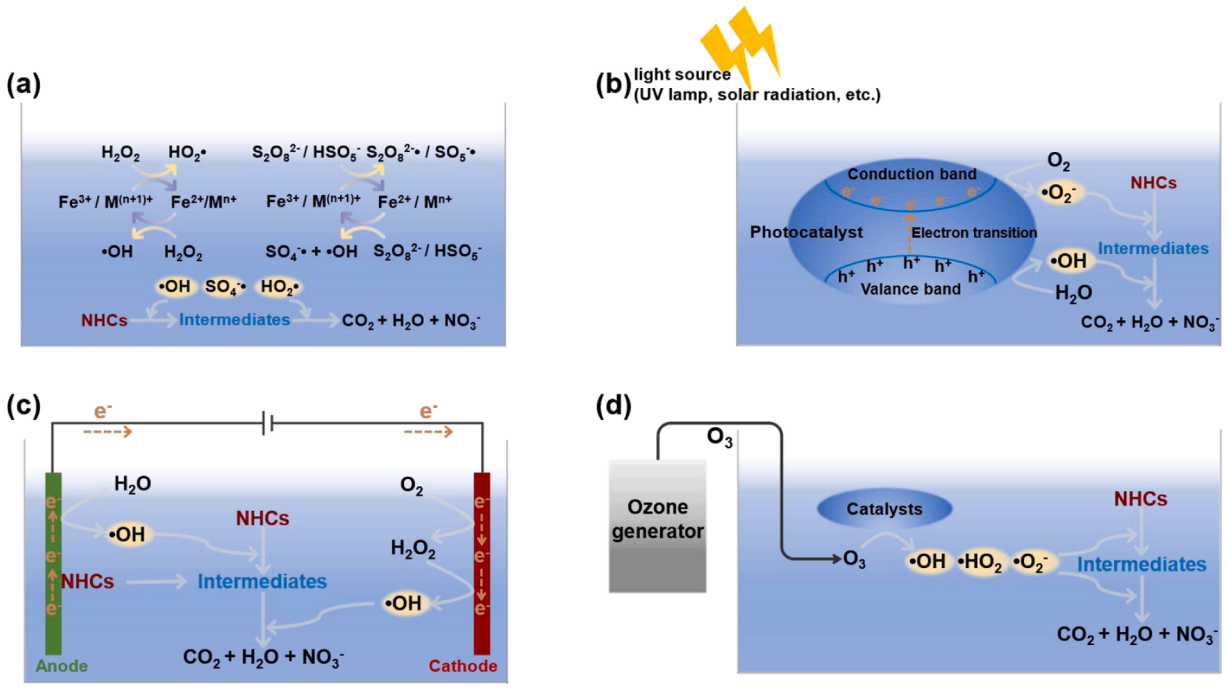


Fig. 3. The degradation mechanisms of NHCs in advanced oxidation processes. (a) Fenton/Fenton-like oxidation, (b) photocatalysis, (c) electrochemical oxidation, and (d) catalytic ozone oxidation.

reduction in 180 min) (Ben Hammouda et al., 2016), while its magnetic nanoparticle-enhanced version achieved better results with 3-methylindole (100 % removal and 80 % TOC reduction in 120 min), with the added benefit of magnetic recovery (Ben Hammouda et al., 2015). Beyond iron, various metal catalysts (Cu^{2+}, Mn^{2+}) and variable-valence metal oxides ($M^{n+}/M^{(n+1)}$) promoted reactive species generation (•HO_2, •OH, SO_4^-•, $^1O_2$) from H_2O_2 and persulfates (including peroxydisulfate (PDS) and peroxymonosulfate (PMS)) by Haber-Weiss chain reaction in Fenton-like systems (Fig. 3a) (Jiao et al., 2022; Li et al., 2014; Singh and Lo, 2017; Tong et al., 2023; Wang and Wang, 2018; Wu et al., 2020). Notable examples included: In H_2O_2 system, copper-based waste printed circuit boards as heterogeneous catalysts realized 90 % pyridine removal and 46.18 % mineralization (Li et al., 2014), hierarchical metal oxides (Mn_3O_4, Co_3O_4, NiO) achieved 90–95 % pyridine removal and 80–85 % mineralization (Singh and Lo, 2017), and CuCe oxide accomplished 98.6 % quinoline degradation and 90.7 % mineralization (Jiao et al., 2022); in persulfates system, Fe_3O_4-doped biochar, which was synthesized from the mixture of Fenton sludge and municipal sludge, realized 97.6 % triazole removal and 80.33 % mineralization (Tong et al., 2023). Comparative study revealed context-dependent effectiveness: While Fenton system outperformed Fe^{2+}-activated persulfate in synthetic wastewater (100 % vs 73.4 % Trimethoprim removal and 54.9 % vs 40.5 % mineralization), persulfate systems showed superior performance in real wastewater (43.6 % vs 35.8 % Trimethoprim removal) due to higher SO_4^-• selectivity (Wang and Wang, 2018). Advanced applications, such as nano zero-valent iron activating both persulfate and H_2O_2, achieved 96 % sulfamethoxazole degradation within 30 min at optimal ratios (Wu et al., 2020).

(2) Photocatalysis

The photocatalytic degradation of NHCs was initiated by photocatalyst activation through appropriate electromagnetic radiation (ultraviolet or solar). This activation induced electronic transition from the valence band to the conduction band, generating photoinduced electron-hole pairs (e^- - h^+). The subsequent charge carrier formation facilitated the production of various ROS (•OH, •HO_2, •O_2^-), which mediated the oxidative degradation and mineralization of NHCs (Fig. 3b). The fundamental mechanisms were as follows (Eqs. (4)–(9)) (Mahajan et al., 2019):

$$Photocatalyst \rightarrow e^-_{cb} + h^+_{vb} \quad (4)$$

$$e^-_{cb} + O_2 \rightarrow •O_2^- \quad (5)$$

$$h^+_{vb} + OH^- \rightarrow •OH \quad (6)$$

$$•O_2^- + H_2O \rightarrow HO_2• + OH^- \quad (7)$$

$$HO_2• + H_2O \rightarrow H_2O_2 + •OH \quad (8)$$

$$•OH + •O_2^- + HO_2• + NHCs \rightarrow CO_2 + H_2O + NO_3^- \quad (9)$$

Photocatalytic systems encompassed diverse materials, including metal oxide semiconductors (TiO_2, ZnO), metal sulfides (ZnS, CdS) (Gupta et al., 2020; Jing et al., 2012; Mahajan et al., 2019; Senasu et al., 2021; Sharma and Kalita, 2025), and metal-ion modified non-metallic materials (graphene, C_3N_4) (Sharma et al., 2020). Among single-component semiconductors, TiO_2 demonstrated particular prominence. TiO_2 nanoparticles achieved 91.5 % quinoline degradation under ultraviolet (UV) irradiation (Jing et al., 2012), while TiO_2 systems utilizing tungsten filament UV sources demonstrated 97 % degradation efficiency for quinoline yellow (Gupta et al., 2012), albeit without reported mineralization rates. Multi-component systems generally exhibited enhanced photocatalytic performance. Cu-doped ZnO catalysts, synthesized via precipitation method, achieved mineralization rates of 92.4 % and 74.3 % for pyridine and quinoline, respectively, under optimized conditions (Mahajan et al., 2019). Similarly, ZnO-TiO_2 mixed metal oxide systems demonstrated 92 % quinoline degradation and 78 % TOC removal after 240 min under optimal conditions (Gupta et al., 2020). Recent advances in composite photocatalysts showed further improvements in catalytic efficiency. V_2O_5-ZnS composites achieved 96 % ciprofloxacin degradation within 150 min, significantly outperforming individual V_2O_5 (77 %) and ZnS (56 %) components (Sharma and Kalita, 2025). Additionally, reduced GO@mesoporous ZnO demonstrated exceptional efficiency, achieving 99 % ofloxacin degradation within 300 min under optimized conditions (Sharma et al.,

2020).

(3) Electrochemical oxidation

Electrochemical oxidation processes utilized electrical energy to facilitate pollutant removal through electrode-mediated reactions, primarily via Anodic Oxidation (AO) and Electro-Fenton (EF) mechanisms (Fig. 3c). The AO process encompassed multiple oxidation pathways: direct electron transfer at electrode interfaces, indirect oxidation via electrochemically generated oxidants (predominantly •OH) in bulk solution, or concurrent operation of both mechanisms. In EF process, continuous H_2O_2 generation occurred at the cathode through two-electron oxygen reduction reaction ($2e^-$ ORR) of supplied oxygen (atmospheric or pure). The generated H_2O_2 subsequently reacted with catalytic Fe^{2+} to produce •OH, facilitating complete mineralization of NHCs to CO_2, H_2O, and NO_3^- (Eqs. (10)–(13)) (Zhu et al., 2020).

$$O_2 + 2H^+ + 2e^- \rightarrow H_2O_2 \quad (10)$$

$$H_2O_2 + Fe^{2+} \rightarrow \bullet OH + OH^- + Fe^{3+} \quad (11)$$

$$Fe^{3+} + e^- \rightarrow Fe^{2+} \quad (12)$$

$$\bullet OH + NHCs \rightarrow CO_2 + H_2O + NO_3^- \quad (13)$$

The efficacy of electrochemical oxidation for NHCs removal was predominantly governed by the catalytic activity of electrodes. In AO process, high-performance catalytic materials, notably boron-doped diamond (BDD), facilitated enhanced electron transfer at electrode interfaces. Xing et al. (2012) demonstrated differential degradation kinetics among NHCs on BDD electrodes, establishing the reactivity sequence: indole > quinoline > isoquinoline > benzotriazole > benzimidazole. Wang et al. (2016a) elucidated the quinoline degradation pathway on BDD anodes, where hydroxyl radical-mediated pyridine ring cleavage generated phenylacetaldehyde, subsequently oxidizing to phenylacetic acid before transformation into short-chain carboxylic acids and final mineralization to CO_2, H_2O, and NO_3^-. In EF process, cathode material optimization focused on promoting two-electron oxygen reduction reaction ($2e^-$ ORR) while suppressing four-electron oxygen reduction reaction ($4e^-$ ORR) to enhance H_2O_2 generation. Chang et al. (2024) developed cobalt-loaded nitrogen-doped porous carbon cathodes exhibiting superior hydrophilicity (45° water contact angle), achieving 89 % H_2O_2 selectivity through enhanced dissolved oxygen interaction. Under optimal conditions, this system achieved complete quinoline degradation within 15 min and 72 % mineralization after 2 h. Synergistic coupling of AO and EF processes demonstrated enhanced remediation efficiency. Santos et al. (2020) employed a BDD anode with modified carbon felt cathode (carbon black/polytetrafluoroethylene composite) for clopyralid degradation. The coupled system achieved 99 % degradation and complete mineralization within 4 and 8 h, respectively, significantly outperforming the singular AO process (4-hour degradation ratio: 58 %, 8-hour mineralization ratio: 65 %). Moreover, recent studies are looking into electrochemical oxidation system incorporating persulfate activation. He et al. (2025) evaluated the removal efficiency of 50 mg/L pyridine using electrochemical oxidation with different oxidants (PDS, PMS, and H_2O_2). The PMS system achieved over 90 % removal within one hour, significantly outperforming PDS and H_2O_2 system, which only achieved approximately 40 % removal. This superior performance is attributed to the ability of PMS system to generate higher concentration of SO_4^-• radical with faster activation kinetics, maintaining effectiveness even under high Cl^- conditions. The PMS system also demonstrated optimal performance in treating real coal wastewater, achieving the lowest UV-absorption values post-treatment.

(4) Catalytic ozone oxidation

Ozone oxidative degradation of NHCs proceeded via two primary pathways: (1) direct molecular ozone attack through electrophilic insertion at electron-rich moieties, though this pathway alone proved insufficient for complete mineralization, necessitating catalytic enhancement; and (2) catalyst-mediated ozone decomposition generating ROS (predominantly •OH and $•O_2^-$) through chain reaction mechanisms, facilitating enhanced oxidative degradation and mineralization (Fig. 3d) (Nawrocki and Kasprzyk-Hordern, 2010). The detailed mechanisms were as follows (Eqs. (14)–(20)) (Liu et al., 2019; Liu et al., 2022a):

$$O_3 + H_2O \rightarrow 2\bullet OH + O_2 \quad (14)$$

$$O_3 + OH^- \rightarrow HO_2^- + O_2 \quad (15)$$

$$O_3 + HO_2^- \rightarrow \bullet OH + •O_2^- + O_2 \quad (16)$$

$$2\bullet OH \rightarrow H_2O_2 \quad (17)$$

$$H_2O_2 \rightarrow HO_2^- + H^+ \quad (18)$$

$$O_3 + H_2O_2 \rightarrow \bullet OH + \bullet HO_2 + O_2 \quad (19)$$

$$\bullet OH + \bullet HO_2 + NHCs \rightarrow CO_2 + H_2O + NO_3^- \quad (20)$$

Catalyst efficiency represented a critical determinant in ozone oxidation of NHCs. Metal oxides (MgO, TiO_2, MnO_2, Al_2O_3, etc.) predominated as heterogeneous catalysts in catalytic ozone oxidation systems (Zhu et al., 2017a, 2017b). Zhu et al. (2017b) demonstrated that Nano-MgO significantly enhanced quinoline degradation efficiency from 53.8 % (sole ozone oxidation) to 90.7 % under identical conditions, primarily through promoted ozone decomposition to •OH, facilitating mineralization. Catalyst recyclability presented a crucial consideration in practical applications. Recent advances incorporated magnetic material spinel-type ferrites supported on natural sepiolite (SEP) to enhance magnetic recoverability. Liu et al. (2019, 2022a) developed $CuFe_2O_4$/SEP and $NiFe_2O_4$/SEP heterogeneous catalysts for quinoline degradation. Both systems demonstrated enhanced ozone and H_2O_2 decomposition, promoting ROS generation (predominantly •OH and $•O_2^-$), thereby accelerating quinoline degradation and mineralization rates. However, system performance exhibited sensitivity to common anions. Specifically, CO_3^{2-}, HCO_3^-, and PO_4^{3-} diminished catalytic activity through active site competition, with PO_4^{3-} additionally functioning as a •OH scavenger.

(5) Supercritical Water Oxidation

Supercritical Water Oxidation (SCWO) utilized supercritical water ($P \geq 22.1$ MPa, $T \geq 374.15$ °C) as a reaction medium to facilitate homogeneous oxidation of NHCs to CO_2, H_2O, and N_2 (Al-Duri et al., 2016; Gong et al., 2016). The primary mechanisms were as follows (Eqs. (21)–(25)) (Al-Duri et al., 2016),

$$H_2O + O_2 \rightarrow \bullet HO_2 + \bullet OH \quad (21)$$

$$\bullet HO_2 + \bullet HO_2 \rightarrow O_2 + H_2O_2 \quad (22)$$

$$H_2O_2 \rightarrow 2\bullet OH \quad (23)$$

$$\bullet OH + H_2O_2 \rightarrow \bullet HO_2 + H_2O \quad (24)$$

$$\bullet OH + \bullet HO_2 + NHCs \rightarrow CO_2 + H_2O + N_2 \quad (25)$$

SCWO offered distinct advantages including rapid kinetics, high conversion efficiency, environmental-friendly products, and economical oxidant utilization. Gong et al. (2016) demonstrated that increasing the oxidation coefficient (0 to 4.0) enhanced 5000 mg/L quinazoline mineralization from 41.4 % to 97.2 % within 7 min with elevated temperatures and extended times promoting complete oxidation of CO to CO_2. Intensification strategies have focused on co-oxidation mechanisms utilizing auxiliary alcohols (Al-Duri et al., 2016; Cocero et al., 2000). These compounds generated additional radical species (•OH,

•HO_2, •CH_3) and provided internal heat through exothermic oxidation reactions, reducing external energy requirements while enhancing NHCs mineralization efficiency (Al-Duri et al., 2016). Al-Duri et al. (2016) demonstrated this approach using isopropanol as a co-fuel with H_2O_2 oxidant for DBU (1,8-Diazabicyclo[5.4.0]undec-7-ene) degradation, achieving 12 % increased mineralization and 34 % enhanced total nitrogen removal.

(6) Microwave and ionizing radiation

Microwave radiation with oxidants (oxygen/air) could indirectly generate •OH, as shown in Eqs. (26) and (27) (Quan et al., 2007):

$$O_2 \rightarrow \bullet O + \bullet O \quad (26)$$

$$\bullet O + H_2O \rightarrow 2\bullet OH \quad (27)$$

which then oxidatively degraded NHCs. This technology could achieve 94 % removal of 1000 mg/L pyridine within 5 min, significantly outperforming traditional catalytic methods such as Ce/SBA-15/H_2O_2 (77 % removal of 800 mg/L pyridine in 5 h) (Subbaramaiah et al., 2013; Zalat and Elsayed, 2013).

Ionizing radiation simultaneously generated both oxidizing (such as ·OH) and reducing agents (such as hydrated electrons, e^-_{aq}) (Wang and Wang, 2007), offering efficient NHCs removal. He et al. (2022) demonstrated this efficiency using ^{60}Co radiation for indole degradation (10 mg/L), where •OH, e^-_{aq}, and hydrogen radicals (H•) collectively achieved 99.2 % removal and 50 % mineralization within 12 min, with the contribution of each radical confirmed by quenching experiments.

The reviewed AOPs predominantly utilized •OH radicals, achieving NHC removal efficiencies exceeding 80 % within 5 h. Microwave radiation, ionizing radiation, and supercritical water oxidation demonstrated superior removal rates with shortest reaction times. However, these processes incurred substantial operational costs, primarily due to power and chemicals consumption. For instance, ionizing radiation (by electron beam) for pharmaceutical wastewater treatment costed up to 4.30 US$/$m^3$. Even the relatively economical electro-Fenton process required 1.82–2.54 US$/$m^3$ (corresponding to 17.5–24.4 kWh/m^3) (Zhu et al., 2022b). Additionally, •OH radicals were readily deactivated by inorganic anions (particularly Cl^-) and high concentrations of matrix organics, limiting their application to relatively low NHC concentrations (<800 mg/L pyridine, <120 mg/L indole, and <200 mg/L quinoline) compared to typical industrial wastewater concentrations (Table S1). As a contrast, SO_4^-• radical exhibited higher selectivity towards NHCs, superior anti-interference capabilities, and better removal performance while reducing chlorinated byproduct formation (He et al., 2025; Wang and Wang, 2018). Notably, SO_4^-• radical-dominant systems demonstrated lower energy consumption (0.56 vs 1.4 kWh/g-pyridine removed for electrochemical oxidation with persulfate vs H_2O_2 activation) while achieving better removal efficiency (He et al., 2025). These advantages suggested that future AOP research should focus on persulfate systems and other multi-reactive species systems.

4.2. Advanced reduction processes (ARPs)

ARPs, a combination of activation technologies and reductants, could generate reductive radicals and efficiently degrade target pollutants. The reported reductive radicals include e^-_{aq}, H•, and sulfite radical (SO_3^-•) (Xiao et al., 2024). Sulfite-UV process was a typical ARPs which has been widely applied in refractory organics treatment, and its performance on NHCs degradation has also been studied. The sulfite-UV system generated reductive radicals mainly through the following reactions (Eqs. (28) and (29)) (Xiao et al., 2024):

$$SO_3^{2-} + h\nu \rightarrow SO_3^-\bullet + e^-_{aq} \quad (28)$$

$$HSO_3^- + h\nu \rightarrow SO_3^-\bullet + H\bullet \quad (29)$$

and the degradation of NHCs in this system was contributed by direct photolysis and deep reduction. The mechanism of direct photolysis was destabilizing the π-bonds of the NHCs and placing them in an excited state. These excited NHCs could react with H_2O and reductive free radicals (e^-_{aq}, H•, and SO_3^-•) to promote their degradation (Liu et al., 2022b). Compared to the UV-H_2O_2 process, Liu et al. (2022b) discovered that the sulfite-UV system significantly enhanced the degradation of pyridine (the pseudo-first-order reaction rate constant was 1.3 times that of the UV-H_2O_2 process), indicating that pyridine might be more effectively removed by reductive free radicals than by oxidative free radicals.

5. Microbial treatment of NHCs

While physicochemical treatment methods for NHC-containing industrial wastewater faced limitations due to high energy costs, complex matrix effects (high salinity, organic content, competing ions), and variable wastewater quality, the microbial treatment offered a cost-effective, versatile, and adaptable alternative. The microbial degradation of typical NHCs followed sequential pathways (Fig. 4) involving initial hydroxylation, followed by carboxylation or amination, and ultimately ammonification and ring-opening to form inorganic-nitrogen and small organic molecules that enter the tricarboxylic acid cycle for complete mineralization (An et al., 2024; Bai et al., 2015; Bankole et al., 2017; Gao et al., 2023; Hu et al., 2019; Padoley et al., 2008; Wang et al., 2018; Zheng et al., 2024). However, the inhibitory effects of NHCs on microbial metabolism necessitated strategies to enhance microbial resilience, detoxification capacity, and metabolic efficiency.

5.1. Aerobic treatment

Zhang's research group (Xu et al., 2017; Lu et al., 2019; Zhang et al., 2019; Lu et al., 2020a) elucidated that the initial monooxygenation reactions are critical in the aerobic biodegradation of multiple NHCs. These reactions, catalyzed by monooxygenase enzymes, required molecular oxygen and electron donors (2H) (Xu et al., 2017) (Fig. 5a). While conventional aerobic biological treatment provided sufficient molecular oxygen, the limited availability of electron donors necessitated supplementation through exogenous sources such as sewage treatment residual sludge (Lu et al., 2020a; Zhang et al., 2019) or fallen leaves (Lu et al., 2019). The electron donation efficiency could be enhanced through UV irradiation (Lu et al., 2019) or ultrasonic treatment (Lu et al., 2020a; Zhang et al., 2019). Research further revealed that accumulation of initial monooxygenation products (e.g., 2-hydroxypyridine from pyridine) inhibited reaction progression, requiring a second monooxygenation for further degradation (Zhu et al., 2021a; Zhu et al., 2021b). Chen et al. (2022) demonstrated enhanced biodegradation through selective acclimation of activated sludge communities to both primary substrates and their hydroxylated intermediates.

Aerobic Granular Sludge (AGS) exhibited superior characteristics compared to flocculent activated sludge, including enhanced settling properties, structural integrity, and resistance to toxic shock loads. Its layered structure incorporated diverse microbial niches (aerobic, facultative, and anaerobic), enabling efficient NHCs removal (Liu et al., 2015; Liu et al., 2018; Tomar et al., 2022). The process involved aerobic transformation and NH_4^+ oxidation to NO_3^- in the outer layer, followed by denitrification in the inner anoxic zone using various electron donors, including intermediates of NHCs degradation and intracellular/extracellular polymers (Liu et al., 2015; Liu et al., 2018). Strains with enhanced self-aggregation properties, such as *Phanerochaete* sp. HSD (Wang et al., 2011) and *Rhizobium* sp. NJUST18 (Liu et al., 2015), characterized by mycelia, interconnected fibers, or flagella, facilitated rapid aerobic sludge granulation. These microorganisms could establish primary granulation cores in harsh environments, expediting the development of resistant AGS systems. Beyond conventional inoculation methods, specific NHC-degrading AGS could be cultivated using functional bacterial strains. For instance, AGS cultivated with *Rhizobium* sp.

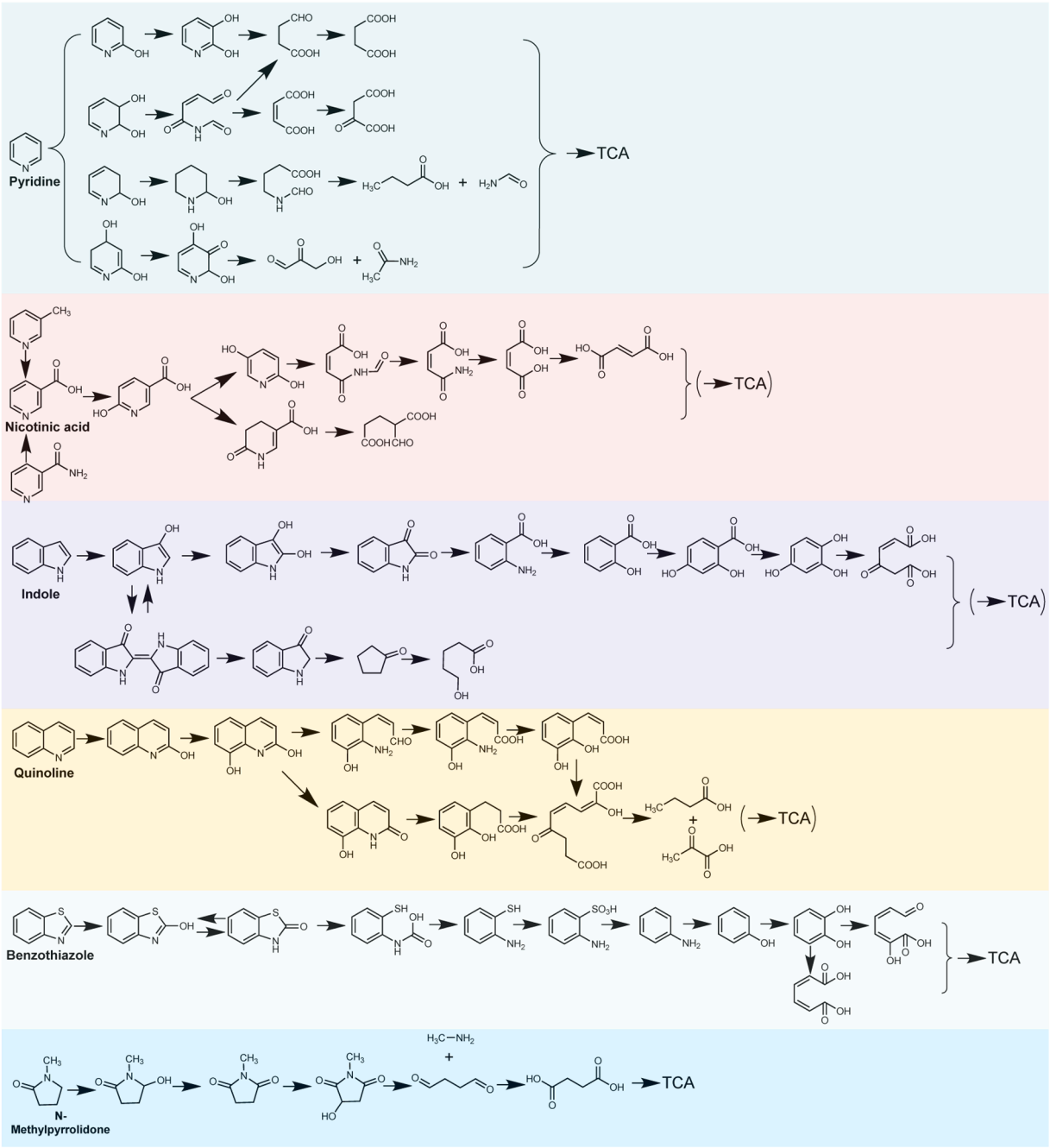


Fig. 4. The degradation pathway of typical NHCs in microbial treatment. The degradation pathways of nicotinic acid, indole and quinoline were reported in both aerobic and anaerobic conditions, with their intermediates entering the tricarboxylic acid (TCA) cycle only under aerobic conditions. The degradation pathways of pyridine, benzothiazole and N-Methylpyrrolidone were only reported under aerobic conditions.

NJUST18, a strain possessing both pyridine degradation capability and self-aggregation properties, achieved nearly complete removal of pyridine (2500 mg/L) within 4 h, whereas pure *Rhizobium* sp. required 250 h for the same treatment. The Rhizobium-cultivated AGS maintained zero-order elimination kinetics despite increasing influent concentrations from 500 to 4000 mg/L (Liu et al., 2015). Complex NHCs, such as indole, presented challenges due to steric hindrance affecting enzymatic binding and potential inhibition of extracellular polymeric substances (EPS) secretion at elevated concentrations. This led to low degradation efficiency and disintegration of AGS (Tomar et al., 2022). Iron-carbon composite supplementation offered a viable solution, promoting EPS secretion for granule stability while enhancing pollutant removal through combined physical-biological adsorption and microbial-AOP degradation (•OH produced from iron-carbon microelectrolysis system), thereby improving system resilience and treatment efficiency (Pan et al., 2024).

5.2. *Anoxic treatment*

Anoxic biological processes offered an energy-efficient alternative to aerobic treatment for NHCs degradation by eliminating aeration requirements. The denitrification process in anoxic systems provided an electron acceptor (nitrate) with redox potential second only to O_2, facilitating organic matter oxidation. Researches demonstrated complete degradation of several recalcitrant NHCs (pyridine, indole, quinoline, isoquinoline, 2-methylquinoline, imidazole, and carbazole) singly

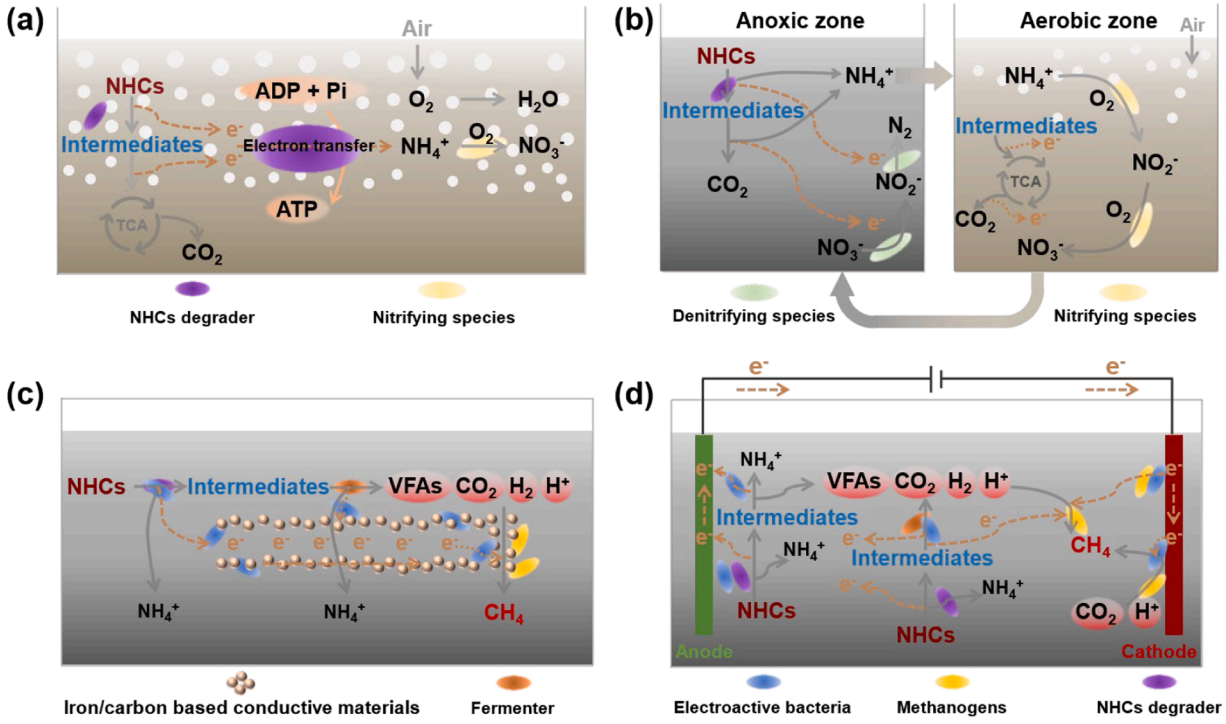


Fig. 5. The degradation mechanisms of NHCs in microbial treatment. (a) aerobic process, (b) anoxic-aerobic process, (c) anaerobic process enhanced by carbon-/iron-based conductive materials, and (d) anaerobic process enhanced by electrical stimulation.

at 50–100 mg/L within 60 h using nitrate as an electron acceptor. Meanwhile, their degradation pathways and mechanisms were analyzed by physicochemical properties (charge density and solubility) (Li et al., 2001), intermediates species, and active sites (PM3 semiempirical quantum chemical calculation) (Xu et al., 2015).

While nitrate supplementation in anoxic systems showed promise, the integration of anoxic-aerobic processes presented a more sustainable approach for enhanced NHCs mineralization (Fig. 5b). This integration eliminated the need for chemical additives while enabling synergistic interactions between the two processes, thereby optimizing overall treatment efficiency. Studies established anoxic/aerobic membrane bioreactors (MBR) and confirmed that nitrate significantly improved NHCs biodegradation through a cyclical mechanism: NHCs underwent ammonification in the anoxic reactor, followed by complete nitrification to NO_3^- (and bio-oxidation of NHCs intermediates) in the aerobic MBR (Hou et al., 2018; Shen et al., 2015). The recycled NO_3^- served as an electron acceptor in the anoxic zone, accelerating NHCs biodegradation (Shen et al., 2015). Notably, certain NHCs, such as N-Methyl pyrrolidone (NMP), could serve as electron donors for exogenous NO_3^- denitrification beyond their stoichiometric requirements (Chen et al., 2024).

Besides integrating anoxic-aerobic systems, recent developments also included the Algal-bacterial Symbiotic System (ABS), which leveraged metabolic synergy between algae and bacteria (Shi et al., 2020b; Zhang et al., 2023). Algae utilized CO_2 and nutrients (NH_4^+, PO_4^{3-}) through photosynthesis, generating O_2 for microbial decomposition of NHCs, which in turn produced CO_2 and nutrients mentioned above, supporting algal growth (Shi et al., 2020b). Shi et al. (2020b) demonstrated superior removal efficiencies for quinoline and indole (99.95 % and 99.34 %, respectively) in a polyurethane-Chlorella-bacteria system compared to single-organism systems (in anoxic bacterial system: 81.77 % and 89.66 %, respectively; in Chlorella system: barely degradable). Zhang et al. (2023) reported complete degradation of pyridine (150 mg/L) and >80 % total nitrogen removal in an ABS without external aeration, highlighting its potential for sustainable wastewater treatment.

5.3. Anaerobic treatment

Anaerobic digestion could be implemented as either a two-phase or single-phase process. Two-phase systems began with hydrolysis-acidification, where ammonification and volatile fatty acids (VFAs) generation occurred, converting NHCs into intermediates without achieving complete mineralization or methane production. This initial phase functioned primarily as a pretreatment for the subsequent methanogenic phase. In single-phase systems, NHC hydrolysis and methane production occurred simultaneously within the same environment. Although anaerobic digestion offers advantages in resource recovery and energy generation, its application to NHC treatment presents significant operational challenges. The inherent sensitivity and slow growth of anaerobic microorganisms, especially methanogens, made them vulnerable to process inhibition from the toxicity of NHCs and intermediates, NH_4^+ accumulation with NHC ammonification (Ronan et al., 2021), and organic acids buildup with NHC hydrolysis-acidification. These limitations necessitated the development of enhancement strategies for stable anaerobic digestion.

5.3.1. Co-digestion

Co-digestion enhanced anaerobic digestion of NHCs through multiple mechanisms: dilution of inhibitory compounds, improved nutrient balance, enhanced buffering capacity, metabolic coupling and increased microbial diversity. These synergistic effects activated or yielded function-specific enzymes, for instance, the growth of syntrophic microbes (e.g. pyruvate-ferrodoxin oxidoreductases, carbon monoxide dehydrogenase, and pyruvate formate-lyase) and methanogens (e.g. coenzyme M methyltransferase and F430—the cofactor of methyl-coenzyme M reductase) (Mu et al., 2020), creating more stable and efficient digestion systems (Karki et al., 2021). Common co-substrates included readily degradable, high-energy organic compounds such as glucose, cellulose, short-chain fatty acids, alcohols, microalgal biomass (Shi et al., 2020a; Karki et al., 2021), and food wastes (Karki et al., 2021).

While most co-substrates reduced the protein/polysaccharide ratio

in EPS and hence the particle size of sludge flocs/granules, potentially compromising anaerobic sludge settling properties, they significantly enhanced microbial community diversity and promoted NHCs hydrolysis to short-chain hydrocarbons (Shi et al., 2019c). For example, carboxymethylcellulose addition improved quinoline and indole removal rates from 85.88 % and 80.34 % to 95.23 % and 94.33 %, respectively

5.3.2. Addition of carbon-/iron-based conductors

Recent advances in anaerobic degradation of NHCs have focused on enhancement through carbon-based and other conductive materials, particularly exploiting direct interspecies electron transfer (DIET) (Fig. 5c). DIET facilitated electron transfer between NHC-degrading bacteria and methanogens through two primary pathways: 1) direct microbial contact via conductive pili, cytochrome c and other membrane-bound proteins, and 2) conduction through conductive materials (Summers et al., 2010). Conductive materials, including biochar, powdered activated carbon (PAC), conductive polymers-modified polyurethane nanocomposites, and conductive/semiconductive iron oxides and oxyhydroxides, enhanced DIET processes and functional microbial enrichment, leading to improved methylation and nitrogen-heterocycle/benzene ring-opening reactions while simultaneously increasing methane conversion efficiency and microbial community stability (Li et al., 2019; Shi et al., 2019a; Shi et al., 2019b; Shi et al., 2019d; Yang et al., 2024b). Li et al. (2019) reported increased abundances of key NHC-degrading bacteria (*Levilinea, Longilinea, Sphingobium*, and *Comamonas*) following conductive material addition (polyurethane, PAC, biochar). Among them, *Sphingobium* harboured gene clusters of *car* (degrading carbazole or indole to anthranilate), *ant* (converting anthranilate to catechol), and *cat* (degradation of catechol), which were essential for complete NHCs degradation (Wang and Xu, 2019). Wang and Xu (2019) also reported similar gene clusters in *Pseudomonas, Sphingomonas*, and *NovoSphingobium* in carbazole system. Notably, *Comamonas* produced nicotinate dehydrogenase, which catalyzed the initial hydroxylation step in microbial NHCs degradation pathways (Lu et al., 2020b). Yang et al. (2024b) demonstrated that while biochar and GAC primarily enhanced pyridine removal through adsorption (95–97 % removal, 24.9–26 % degradation), magnetite showed superior performance in both degradation (28 %) and methane production (87.6–94.4 vs 97.6 mL/d).

Composite conductive materials exhibited superior enhancement of anaerobic NHCs degradation compared to single conductive materials, as demonstrated by Han's research group (Li et al., 2019; Shi et al., 2019a; Shi et al., 2019b; Shi et al., 2019d) through the integration of iron oxides and oxyhydroxides with conductive carbon substrates (Fe$(OH)_3$@PAC, Fe$(OH)_3$@biochar, Fe_3O_4@polyurethane). This composite design not only improved electron transfer efficiency but also effectively prevented powdered/nanoparticulate iron loss during long-term reactor operation. For instance, the enhanced performance was evidenced by Fe_3O_4@polyurethane achieving significantly higher removal efficiencies for pyridine (80.32 %), indole (98.24 %), and quinoline (93.22 %) at 100 mg/L compared to unmodified carriers (50.33 %, 90.22 %, and 91.22 %, respectively) under 48-hour hydraulic retention time (HRT) (Shi et al., 2019b). Additionally, the released Fe^{3+} promoted microbial aggregation through surface charge reduction and EPS binding (Shi et al., 2019d).

Carbon-/iron-based conductive materials could also facilitate electron transfer to external circuits in microbial fuel cells (MFCs), enabling direct conversion of NHCs chemical energy to electrical energy. Luo et al. (2010) achieved 88 % indole removal (250 mg/L) within 3 h, generating 2.1 W/m^3 maximum volumetric power density. Jian et al. (2020) further enhanced MFC performance using a nitrogen-doped carbon felt anode with Fe_2O_3@polyaniline-dopamine film. This modified anode was more hydrophilic and conductive, achieving doubled power density (3184.4 ± 45 mW/m^2) and improved indole removal (90.3 % vs 44 % with conventional carbon felt anode).

5.3.3. Electro-assisted anaerobic digestion

Electrical stimulation has emerged as a promising approach for enhancing anaerobic degradation of NHCs. This method employed external voltage supply in anaerobic digestion systems, promoting electroactive biofilm formation on electrode surfaces and improving oxidative degradation efficiency on the anode (Gao et al., 2020; Jiang et al., 2018) (Fig. 5d). The voltage (typically below 1.2 V) facilitated electron transfer from bioanode (NHCs oxidized) to (bio)cathode (electron-acceptor reduced, e.g., methane generation), and meanwhile, this voltage limitation prevented excessive electrode potential or current that could inhibit microbial metabolism and growth (Ding et al., 2016).

The electro-assisted anaerobic digestion system could create a microaerobic environment within the bioanode, which might facilitate the enrichment of various functional microbes and, hence, reinforce system stability under high NHCs loads (Jiang et al., 2018). Studies on quinoline degradation revealed increased abundances of functional bacteria, including *Pseudomonas* (NHC-degrading bacteria), *Desulfovibrio* (electroactive bacteria), *Geobacter* (electroactive bacteria), *Methanocorpusculum* (methanogen), *Comamonas* (denitrifying bacteria), *Planctomycetes* SM1A02 (anaerobic ammonia-oxidizing bacteria) and *Nitrosoarchaeum* (ammonia-oxidizing archaea) (Gao et al., 2020). Notably, *Pseudomonas*'s functional genes (*qorL, oxoO*, and *oxoR*) encoded monooxygenases capable of transforming quinoline into hydroxylated intermediates (2-hydroxyquinoline and 2,8-dihydroxyquinoline) (Sun et al., 2009). Jiang et al. (2018) reported enhanced pyridine biodegradation using graphite-felt bioelectrodes, achieving 100 % removal and 96.1 % mineralization. The anodic biofilm was enriched with key functional bacteria: *Paracoccus*, which harboured N-methylhydantoin amidohydrolase (*hyuB*) for NHCs ring-opening (Wang et al., 2019); electroactive bacteria *Desulfovibrio*; and denitrifying bacteria *Dokdonella* and *Hydrogenophaga*. The limited NH_4^+ conversion (60 %) coupled with significant nitrogen removal (40 %) suggested alternative nitrogen removal pathways facilitated by these denitrifying populations.

The electro-assisted system also exhibited enhanced stress tolerance, maintaining efficient NHCs removal under challenging conditions such as high salinity and sulfate concentrations. For instance, a bioelectrochemical system applying 0.8 V external voltage successfully degraded 200 mg/L metronidazole in seawater, primarily through the activity of sulfate-reducing bacteria, sulfur-oxidizing bacteria, and hydrocarbon-degrading bacteria (Xin et al., 2024).

While current research predominantly focused on single-phase anaerobic digestion, two-phase systems offered greater feasibility for NHC treatment due to the distinct environmental requirements of NHC-degrading microbes and methanogens. The hydrolysis-acidification phase could operate under higher oxidation–reduction potential (ORP) conditions without requiring strict anaerobiosis, unlike the methanogenic phase. Additionally, each phase could be optimized through different enhancement strategies. The hydrolysis-acidification phase benefited from micro-aeration (Yang et al., 2025; Yang et al., 2024a; Yang et al., 2023; Sun et al., 2020), where oxygen initiates NHC degradation through monooxygenation and serves as an electron acceptor, enhancing NHC oxidation. The methanogenic phase could be improved by promoting DIET as Sections 5.3.2 and 5.3.3 summarized. Notably, these phases need not be physically separated into distinct tanks but can be established through environmental gradients, such as spatial variation in dissolved oxygen concentration (Shi et al., 2024).

5.4. Enhanced microbial treatment with functional bacteria/enzymes

The incorporation of functional bacteria (Table S4)/enzymes (Table S5) offered a promising microbial enhancement strategy for refractory NHCs treatment. The bioaugumentation performance of functional bacteria: *Stenotrophomonas maltophilia* J2 (pyridine) (Niu et al., 2023), *Exiguobacterium aurantiacum* (MTCC 5638) (pyridine and quinoline) (Jeswani and Mukherji, 2015), *Bacillus tropicus, Bacillus aquimaris* and *Rhodococcus ruber* (pyridine and quinoline) (Zhu et al., 2021a,

2021b), *Bacillus* sp. Q2 (quinoline) (Tuo et al., 2012), *Comamonas testosterone* and *Rhodococcus ruber* (quinoline) (Zhu et al., 2023) were validated in synthetic wastewater with over 90 % of degradation within 24 h when treating wide range of NHCs (pyridine < 2000 mg/L, quinoline < 500 mg/L). Especially, *Bacillus* sp. Q2 was added to a pilot-scale anoxic-oxic reactor treating refinery wastewater and still achieved complete quinoline (160 mg/L) removal within 24 h.

Enzyme-catalyzed processes, particularly using laccase (Xue et al., 2020; Jia et al., 2024; Gu et al., 2019) and Horseradish Peroxidase (Liu et al., 2021), provided environmentally compatible and efficient treatment under mild conditions with high catalytic specificity. These enzymes are preferred due to their molecular stability and ease of purification (Liu et al., 2021; Xue et al., 2020). Redox mediators, such as ABTS (2,2′-azino-bis-3-ethylbenzothiazoline-6-sulfonic acid) (Xue et al., 2020; Gu et al., 2019) and HBT (1-hydroxybenzotriazole) (Jia et al., 2024), enhanced enzyme activity by facilitating electron/hydrogen transfer during redox processes (Liu et al., 2021). Practical applications relied on enzyme immobilization using carriers like alginate salts (Xue et al., 2020), magnetic nanoparticles, and MOFs. Co-immobilization of enzymes and mediators on single carriers optimized catalytic performance (Gu et al., 2019). The degradation mechanism involved laccase-mediated oxidation of mediators to radical cations, which drove non-enzymatic NHCs oxidation while undergoing reduction to their original state (Xue et al., 2020). The bioaugumentation performance of all the functional enzymes were validated in synthetic wastewater with complete degradation within 24 h when treating low concentration of NHCs (< 100 mg/L).

6. Combined treatment strategies for NHC removal

AOPs were commonly employed as pretreatment methods to partially detoxify NHCs before microbial treatment, addressing their inhibitory effects on microbial metabolism and growth. Both sequential and synchronous combined treatment approaches have been developed, including Fenton oxidation (Padoley et al., 2011), UV irradiation (Zhang et al., 2014), catalytic ozone oxidation (Zhu et al., 2017a), or electrochemical redox (Lou et al., 2021) followed by microbial treatment systems, as well as integrated processes such as bio-electro-Fenton systems (Li et al., 2020) (Fig. 6).

Padoley et al. (2011) investigated Fenton oxidation pretreatment for enhancing NHC biodegradability. When treating real chemical industry wastewater containing pyridine (1700 mg/L) or 3-cyanopyridine (725 mg/L), the Fenton process significantly improved biodegradability, as indicated by increased B/C ratios from 0.037 to 0.79 and 0.125 to 0.94, respectively. The Fenton-aerobic combined treatment achieved removal rates of 84 % for pyridine and 99 % for 3-cyanopyridine, with corresponding mineralization rates of 82 % and 93 %. Zhang et al. (2014) reported that UV photolysis pretreatment enhanced pyridine removal rates by 15–43 % (at concentrations 130–310 mg/L) compared to direct aerobic biofilm degradation, with TOC removal increasing by 12 % at 206 mg/L pyridine concentration. In treating coal gasification wastewater, Zhu et al. (2017a) combined multiphase catalytic ozone oxidation using magnesium oxide nanoparticles with an anoxic-aerobic MBR system. This combined approach treating real biologically pretreated coal gasification wastewater achieved complete removal of quinoline (20 mg/L), pyridine (20 mg/L), and indole (10 mg/L), compared to removal rates of 62.5 %, 19 %, and 29 % in the anoxic-aerobic MBR alone. The system also demonstrated high removal efficiencies for COD (92.7 %), NH_4^+ (95.6 %), and total nitrogen (80.6 %). Lou et al. (2021) developed an Electrochemical-Aerobic Fixed Bed Biofilm Reactor for treating multiple pyridine derivatives (100 mg/L pyridine, 3-cyanopyridine, and 3-chloropyridine, respectively). Without electrochemical activation, the system achieved 95 %, 100 %, and 18 % removal rates for pyridine, 3-cyanopyridine, and 3-chloropyridine, respectively. Electrochemical activation (40 mA/cm^2) significantly improved the removal rate of 3-chloropyridine, the most toxic and persistent compound, to 97 %. Li et al. (2020) achieved 94.66 % degradation of 25 mg/L sulfamethoxazole using an MFC-electro-Fenton system with bioanode and CNT-γFeOOH cathode, while realizing a maximum MFC power density of 283.32 mW/m^2.

Fig. 7 presented a statistical analysis of NHC removal efficiencies (focusing on pyridine, quinoline, and indole) across different treatment approaches: AOPs, aerobic biological treatment, anoxic/anaerobic biological treatment, and combined processes (AOP-biological treatment). Microbial processes predominantly handled high-concentration NHCs, whereas AOPs and AOP-microbial combined processes typically treated medium to low concentrations. The HRT followed the order: anaerobic biological > aerobic biological > combined processes > AOPs. Average NHC removal efficiencies ranked as: combined processes > anaerobic biological > AOPs > aerobic biological treatment. Combined treatment processes demonstrated optimal performance while maintaining moderate HRT. Adsorption removal was rarely integrated into combined treatment strategies, likely due to the high costs of adsorbent preparation and modification, complex regeneration requirements, and uncertain long-term stability. However, as Section 3 indicated, adsorption technology demonstrated significant advantages in handling broader and higher concentration ranges compared to other treatments shown in Fig. 7, with equilibrium times comparable to AOP under high concentrations. This capability positions adsorption as a potential buffer or emergency treatment technology for subsequent chemical and

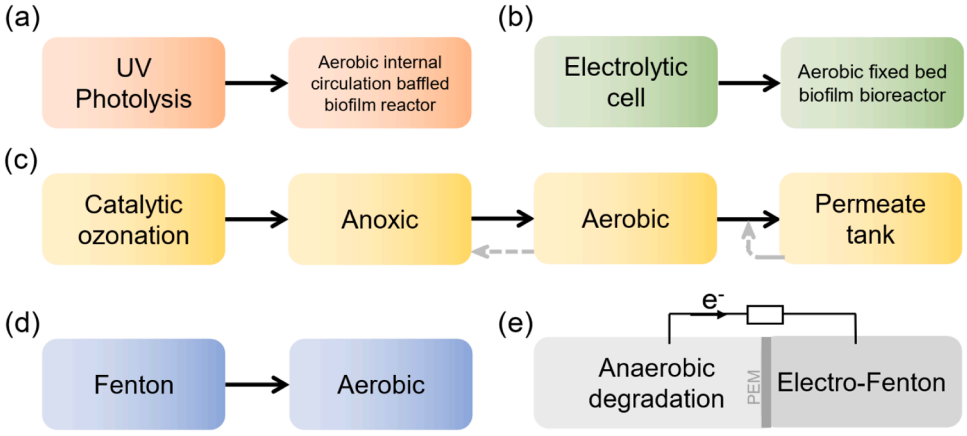


Fig. 6. Combined treatment strategies for NHC removal (a) UV irradiation-aerobic biofilm treatment (Zhang et al., 2014), (b) Electrochemical-Aerobic Fixed Bed Biofilm Reactor (Lou et al., 2021), (c) catalytic ozone oxidation-anoxic-aerobic MBR (Zhu et al., 2017a), (d) Fenton oxidation-aerobic treatment (Padoley et al., 2011), and (e) bio-electro-Fenton systems (Li et al., 2020).

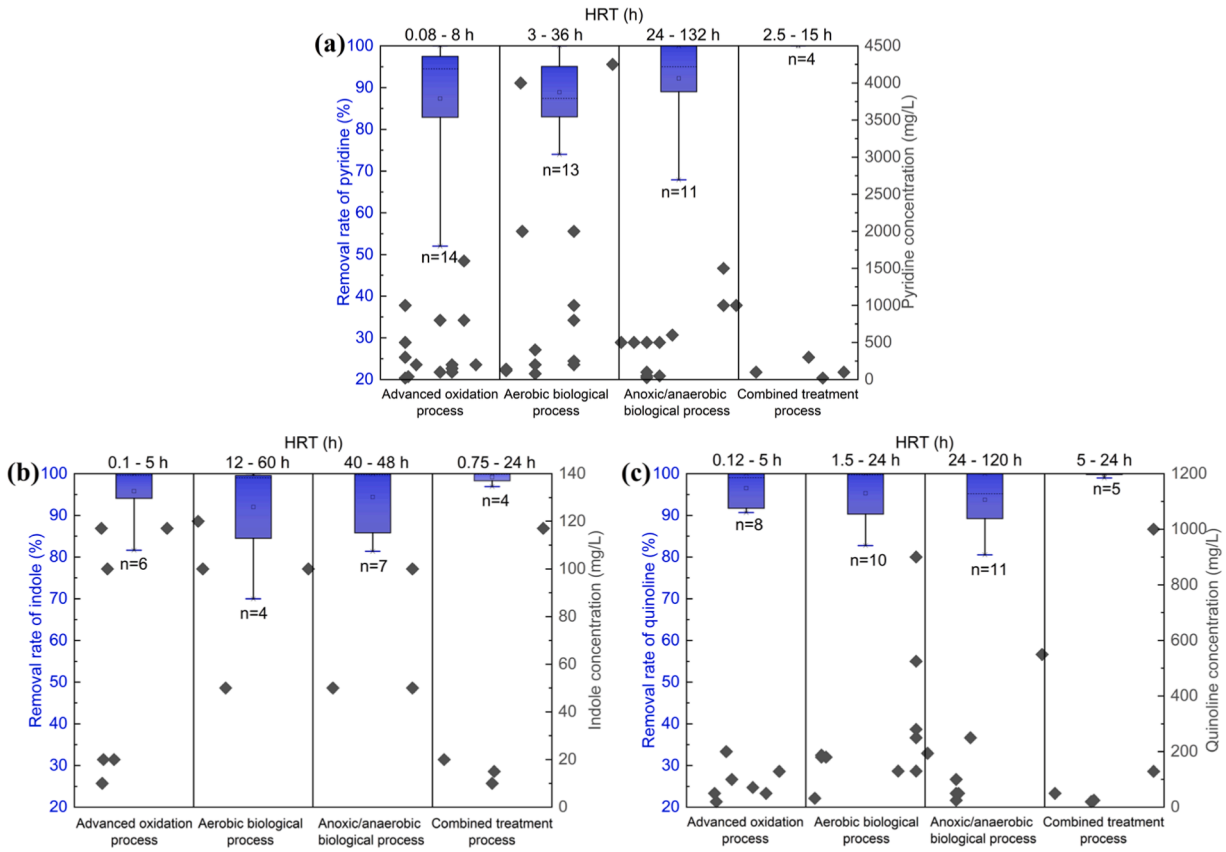


Fig. 7. Removal efficiencies of typical NHCs (a) pyridine, (b) indole, and (c) quinoline in wastewater using advanced oxidation process, aerobic biodegradation, anoxic/anaerobic biodegradation, and combined treatment process; n = number of samples.

biological processes when dealing with irregular industrial wastewater compositions or loading shock scenarios. Additionally, adsorption technology becomes essential as pre-treatment (before microbial treatment) when heavy metal ions are present in wastewater. The development of adsorbents, particularly reticular framework nanoparticles with enhanced water stability and adsorption capacity, remains crucial, with machine learning and large language models emerging as promising tools for material discovery (Zheng et al., 2025).

Current research predominantly emphasizes individual enhancement of adsorption capacity, chemical and microbial degradation efficiency, focusing on specific NHC removal rather than comprehensive treatment strategies and mineralization. The literature reveals significant gaps between laboratory research and practical applications, evidenced by the prevalence of model chemical studies over real industrial wastewater treatment (Zhu et al., 2022b, He et al., 2025; Pan et al., 2024; Chen et al., 2024; Padoley et al., 2011; Zhu et al., 2017a) and the scarcity of pilot-scale investigations (Chen et al., 2024; Cocero et al., 2000) (Table S6). Comprehensive analyses of operational costs and environmental impacts are notably lacking in current literature. These limitations underscore the urgent need to develop integrated treatment processes that effectively and cost-efficiently remove and mineralize NHCs in complex industrial matrices under real-world conditions.

7. Future outlooks towards energy valorization and process design

A comprehensive framework focusing on energy-efficient and environmentally sustainable solutions is required for treating NHC-containing industrial wastewater on an engineering scale. Firstly, the selection criteria for core treatment units should be analyzed, emphasizing the advantages of optimal detoxification and energy valorization. The discussion should then extend to process design considerations, addressing the challenges of managing intermediate products, particularly ammonia and volatile fatty acids (VFAs), and the importance of integrated environmental and economic assessments in process optimization. Finally, the transformative role of machine learning (ML) in enhancing treatment efficiency and process optimization, from pollutants identification and quantification, parameter adjusting guidance to comprehensive life cycle assessments, should be explored. Throughout these aspects, the strategies for future process design emphasize the importance of balancing removal efficiency, energy consumption, and resource recovery while maintaining system stability and adaptability to varying influent characteristics. This holistic approach aims to bridge the gap between experimental findings and practical industrial applications (Fig. 8).

7.1. Core treatment units for NHC-containing industrial wastewater

Given the complex and variable composition of NHC-containing industrial wastewater, high-cost physicochemical strategies (adsorption, AOPs, and ARPs) are unsuitable as core treatment units. NHCs, as cyclic organics rich in carbon and nitrogen, require specific microbial treatment steps: initial ring opening and ammonification to convert organic nitrogen to ammonia, followed by the removal of high-concentration intermediate organics and ammonia. While aerobic treatment can facilitate these steps, its high energy requirements for ring opening, ammonification, and nitrification, coupled with the necessity for anoxic denitrification, make it less favorable. Anaerobic digestion offers significant advantages through reduced energy consumption and the potential of recovering organic carbon as methane, making it a superior choice as the core treatment unit.

However, NHCs toxicity may inhibit methanogens, resulting in low

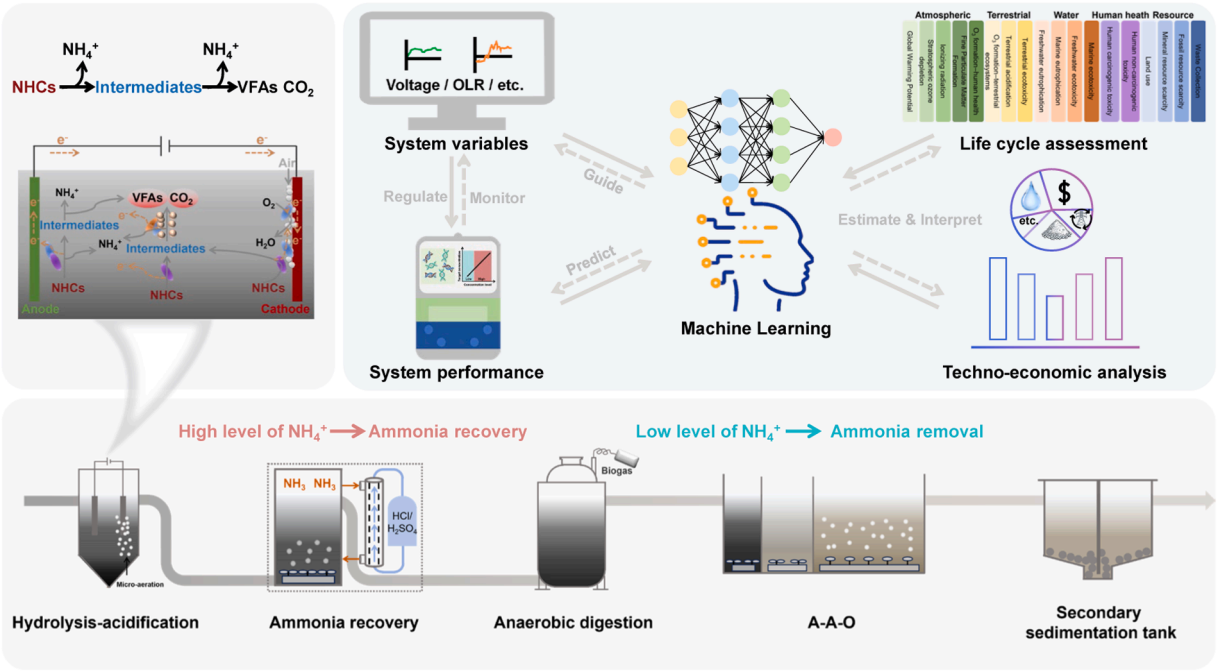


Fig. 8. Proposal of a comprehensive framework focusing on energy-efficient and environmentally sustainable solutions for treating NHC-containing industrial wastewater.

organic loading capacity, poor mineralization rates, and system instability, necessitating pretreatment unit. The ideal pretreatment should reduce toxicity and enhance biodegradability with minimal energy input while preserving electron donors for subsequent energy recovery. Among the three major pretreatment approaches: adsorption merely transfers NHCs and organic carbon rather than transforming them, failing to preserve electron donors; chemical degradation can be controlled to degrade NHCs while retaining most of the electron donors but involves high operational costs and reduced efficiency in complex water matrices due to radical scavenging (although non-radical pathways offer alternatives despite slower kinetics and higher catalyst costs); hydrolysis-acidification microbial treatment emerges as the most suitable pretreatment, achieving pollutant transformation (smaller molecular weight and improved biodegradability) with minimal electron donor consumption, requiring no aeration or chemical additives, maintaining a low carbon footprint, having no strict requirement on anaerobic conditions and demonstrating high feasibility and resilience. This approach can be enhanced through mild microenvironments (microbial carriers) and electron transfer promotion strategies (micro-aeration, conductive materials, electrical stimulation). For example, Shi et al. (2024) reviewed the mechanisms of electrostimulation coupled with micro-aeration around bioanodes for removing electron-withdrawing azo compounds, demonstrating that electrostimulation strengthens electron transfer and microbial associations, while micro-aeration enhances hydrolase and intracellular oxygenase generation. Electrostimulation enhanced the ammonification rate by 15 % and 36 % when treating N, N-dimethylformamide (a dipolar protophilic solvent, similar to pyridine) (Yang et al., 2024a) and pyridine (Jiang et al., 2018), respectively, while promoting the removal rate of the targeting compound by 32 % and 40 % when treating azo compound (Wang et al., 2025b) and pyridine (Jiang et al., 2018), respectively.

Our approach to pre-treating electron-donating NHCs through hydrolysis-acidification utilizes electro-assisted micro-aerobic biocathodes as electron acceptors. Driven by cathodic oxygen reduction and external power, numerous micro-potential differences and weak electrical circuits in the mixed liquor are created to promote refractory organic matter oxidation and detoxification. Precise oxidation–reduction potential (ORP) control ensures preservation of hydrolyzed organic matter (mainly VFAs) for energy recovery and enables ammonia separation/transformation in subsequent units. Therefore, hydrolysis-acidification pretreatment enhanced by electro-assisted micro-aeration, coupled with anaerobic digestion technology (e.g., anaerobic membrane bioreactors), represents an energy-valorized, low-carbon, and resilient approach for simultaneous nitrogen and carbon removal.

7.2. *Treatment process design and optimization*

The design of treatment process for NHC-containing industrial wastewater must consider both raw wastewater toxicity and the impacts of intermediate products on subsequent treatment units. For instance, prior to microbial treatment, it is necessary to evaluate whether the pollutant loading rate exceeds the treatment capacity of existing biological units and to assess the presence of heavy metal ions that may require removal or adjustment through adsorption processes; before the effluent of hydrolysis-acidification tank enters the anaerobic digestion unit, evaluation of ammonia concentration, VFA concentration, and residual pollutants' toxicity is essential, as these factors can inhibit subsequent methane conversion and organic carbon mineralization. For ammonia management, at lower concentrations, biological denitrification processes (e.g., A/O) can be implemented after the anaerobic digestion unit. However, at higher concentrations, ammonia separation becomes necessary before anaerobic digestion to prevent free ammonia inhibition of methanogens. Separation methods include bipolar membranes coupled with membrane contactors or stripping. This approach enables both ammonia recovery and prevention of biological inhibition. Regarding VFAs and toxic substances management, organic loading rate (OLR) serves as the primary control parameter. Effective organic carbon conversion and TOC removal can be achieved through setting appropriate OLRs, effluent recirculation for dilution, and regulation of electron and proton storage/transfer (e.g., Fe_3O_4@Biochar addition (Wang et al., 2023)). As reviewed above, chemical methods typically treat low-concentration NHCs with shorter HRT and better removal/mineralization rates, making them suitable for polishing bio-treated effluent to enhance final water quality.

Besides considering treatment performance and effluent quality,

process optimization requires comprehensive life cycle assessment (LCA) and technical-economic analysis (TEA). For instance, the decision between ammonia removal versus recovery necessitates evaluation of environmental impacts of residual pollutants, environmental benefits of ammonia recovery (compared to the Haber-Bosch process), energy and chemical consumption per ton of water treated, and net economic costs. Ammonia recovery technologies are cost-efficient only when ammonia concentration is high enough. When selecting ammonia removal technologies, environmental impacts vary significantly. Traditional nitrification-denitrification shows high energy consumption and indirect carbon emissions. While partial nitritation/anammox offers lower energy consumption, it risks high N_2O emissions (greenhouse warming potential of N_2O equals 298 CO_2e (IPCC, 2007)), potentially leading to high direct carbon emissions. Feammox represents a potentially low-carbon pathway with successful applications in treating medium-concentration industrial wastewater (Yang et al., 2021).

7.3. Role of ML in treatment units and process optimization

ML serves as a pivotal tool in setting core operational parameters, predicting effluent characteristics, and optimizing process routes. Rapid identification and quantification of primary pollutants, and assessment of their toxicity and biodegradability are crucial for varied industrial wastewater. Research has demonstrated using electroactive biofilm sensors combined with MEA (mind evolutionary algorithm)-ANN (artificial neural network) models for identifying and quantifying multiple toxicants in complex water matrices (Wang et al., 2025a). Besides bioelectricity signals from biosensors, spectral data from *in situ* optical sensors can also serve as monitoring parameters and input variables for ML model development to identify predominant influent pollutants including NHCs and heavy metal ions. Additionally, supervised DBN (deep belief network)-PSO (particle swarm optimization) models enable soft-sensing of difficult-to-monitor biological indicators like BOD_5 based on basic water quality parameters (Zhu et al., 2022a). Following the identification of primary NHC pollutants, ML can be employed to predict the toxicity of NHCs and their intermediates based on structural characteristics, thereby evaluating wastewater detoxification efficiency. This process involves developing Quantitative Structure-Activity Relationship (QSAR) models through ML, utilizing *priori* knowledge of NHC molecular descriptors and their corresponding toxicity data. The established QSAR models can then predict the toxicity of co-existing multiple NHCs and their intermediates, enabling the assessment of overall wastewater detoxification efficiency.

Predicting effluent quality under varying environmental and operational conditions is the guide for parameters adjustment and process optimization. For example, data-driven models that only monitor minimal basic variables have been developed while achieving high-accuracy predictions of A/O effluent quality. These models dynamically update parameters based on instantaneous water quality and process conditions, utilizing multi-objective genetic algorithms to optimize key operational parameters, achieving dynamic balance between energy consumption and effluent quality (Dai et al., 2025). For the proposed pretreatment unit (hydrolysis-acidification with electro-assisted micro-aerobic biocathodes), ML approaches can optimize it by identifying optimal microbial strain combinations for inoculation or bio-augmentation of the hydrolysis-acidification, based on the analysis of microbial population sensitivity to specific NHCs and the biodegradation performance (effluent quality). During operation, ORP (determined by voltage and aeration rate) and OLR might emerge as critical parameters requiring dynamic optimization. ML algorithms can guide the real-time adjustment of these parameters in response to varying influent characteristics by correlating effluent quality with different ORP and OLR conditions. For multi-unit processes, accurate effluent quality prediction requires integration with bioinformatics tools to analyze how environmental and operational conditions affect microbial community structure and interactions, functional genes, and metabolic activity. Research has successfully employed random forests, CNN (convolutional neural networks) with picturized microbiomes, and FCNN (fully-connected neural networks) embedded with microbial interaction networks as priori knowledge to predict microbial composition, interactions, and metabolic activity (Hernández Medina et al., 2022). LSTM (long short-term memory) models have successfully predicted dissolved oxygen concentration, effluent COD, and functional bacterial abundance dynamics in full-scale AAO process wastewater treatment plants (Ma et al., 2025).

ML also enhances TEA and LCA for process route optimization. Under varied influent conditions, real-time comparative analysis of carbon footprints, environmental impacts, and cost-effectiveness of different process routes and operational conditions aids decision-makers in dynamically evaluating potential alternatives (Ghoroghi et al., 2022). This is particularly valuable for resource recovery processes and novel treatment units, where traditional LCA methods face inventory completeness challenges. Supervised ML algorithms (e.g., ANN) could predict inventory and missing data, and rapidly estimate life cycle impacts with improved spatial and temporal explicitness compared to traditional process-simulation approach. Unsupervised learning algorithms assist LCA interpretation through rapid uncertainty and sensitivity analysis, quantifying factor contribution and revealing system key drivers (Romeiko et al., 2024).

When applying ML to wastewater treatment processes, beyond considering data collection/generation and preprocessing, dataset partition, feature selection, model evaluation, and hyperparameter optimization (Zhu et al., 2024), several key considerations emerge: 1) large data volumes are not indispensable (Ma et al., 2025; Zhu et al., 2022a); sufficient data variability is the key. The value of data-driven models lies in extracting specific features of interest to reveal relationships, identify important factors, and quantify their contributions. 2) Integrating prior knowledge, mechanistic models, and data-driven approaches is essential. ML should simplify mechanistic models and eliminate uninformative covariates, while mechanistic models and prior knowledge constrain algorithmic exploration space, forcing data-driven models toward interpretable insights. Especially in microbial applications, the paradigm typically involves innovative experimental methods combined with appropriately abstracted models to evaluate difficult-to-measure parameters and enhance biological relevance (Lopatkin and Collins, 2020). Examples include deriving simplified criteria using SVM (support vector machines) to predict mutualistic bacterial community outcomes (Wu et al., 2019) and quantifying bacterial growth dynamics from metagenomic samples without complete sequence (Gao and Li, 2018). The focus should remain on interpretability rather than pursuing perfect fitting.

8. Conclusions

This review summarized the strengths and limitations of current NHC removal technologies, including adsorption, AOPs/ARPs, and microbial treatments. While adsorption was energy-efficient, it lacked degradation capability and water stability, and AOPs/ARPs, though efficient, were energy/reagent-intensive. Microbial treatments, particularly anaerobic digestion, offered energy recovery potential but were hindered by challenges such as NHC toxicity and methanogenesis inhibition. To overcome these limitations, innovative strategies like conductive materials, electrostimulation and bioaugmentation have been explored. However, most of these technologies remained at the experimental stage of treating synthetic wastewater and faced significant limitations in industrial applications. To bridge the gap between experimental findings and practical industrial applications, this review established a comprehensive framework focusing on energy-efficient and environmentally sustainable solutions for treating NHC-containing industrial wastewater. Firstly, the selection criteria for core treatment units should emphasize the advantages of optimal detoxification and energy valorization. The discussion should then extend to process design

considerations, addressing the challenges of balancing removal efficiency, energy consumption, and resource recovery, incorporating environmental and economic impacts. Finally, employing machine learning in optimizing operational parameters, predicting effluent quality, and supporting LCA & TEA is recommended, for its enabling dynamic process design decisions and energy-resource recovery improvement. This review aimed to inspire innovative research on NHC degradation and conversion, aligning with global sustainability goals and leveraging advancements in AI-driven wastewater treatment technologies.

CRediT authorship contribution statement

Chao Ma: Writing – original draft, Visualization, Investigation, Data curation. **Huiqin Zhang:** Writing – original draft, Formal analysis, Data curation. **Ziwei Liu:** Writing – review & editing, Funding acquisition, Data curation, Conceptualization. **Xinran Meng:** Writing – review & editing, Formal analysis. **Sijia Chen:** Writing – review & editing, Formal analysis. **Jingsong Zhang:** Funding acquisition, Conceptualization. **Yeqiang Li:** Resources, Funding acquisition. **Xia Huang:** Writing – review & editing, Resources, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.watres.2025.123575.

Data availability

Data will be made available on request.

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